

Pelin Kincir

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EDUCATION

- **ETH Zurich** Zurich, Switzerland
M.Sc. in Electrical Engineering and Information Technology
September 2024 - Current
 - Signal Processing and Machine Learning Track
- **Bogazici University** Istanbul, Turkey
B.Sc. in Electrical and Electronics Engineering (4 Years + English Preparatory Year)
September 2019 – July 2024
 - Signal Processing Track

PUBLICATIONS

- S. Bairagi, C. F. Otesteanu, M. Zada, **P. Kincir**, F. Schaub, and C. Menon, “Self-powered Fully Nanofibrous Triboelectric Nanogenerator for Intelligent Knee Joint Angle Monitoring,” submitted to *Advanced Fiber Materials*, 2026.

RESEARCH PROJECTS

- **ARLab: An AI Enabled, AR-Guided Platform for Zero-Shot Wet-Lab Procedures** Boston, MA
Broad Institute of MIT and Harvard - Traverso Lab
July 2026 - January 2027
 - Master’s thesis focused on developing an AI-powered augmented reality (AR) assistant for laboratory procedures using Apple Vision Pro.
 - Integrating computer vision, object tracking, scene understanding, and large language models to recognize laboratory states and provide context-aware guidance without task-specific training.
 - Investigating zero-shot procedural reasoning for real-time assistance, mistake detection, and workflow monitoring in biomedical laboratory environments.
 - **Technical Skills:** Python, Swift, visionOS, Apple Vision Pro, computer vision, object tracking, LLMs/VLMs, AR/VR
- **AI for Joint Angle Monitoring and Movement Classification from TENG Data** Zurich, Switzerland
ETH Zurich - BMHT Lab
March 2026 - June 2026
 - Developed machine-learning methods for joint-angle regression, movement classification, and gait recognition from triboelectric nanogenerator (TENG) knee-sensor signals.
 - Designed signal-processing and feature-extraction pipelines and evaluated intra-participant and leave-one-participant-out generalization using TabPFN-based models.
 - Investigated participant-specific signal normalization strategies to improve cross-subject robustness and analyzed sensor artifacts and inter-subject variability.
 - Research contributed to a manuscript submitted in 2026.
 - **Technical Skills:** Python, PyTorch, Signal Processing, Machine Learning
- **Scene Flow Estimation from Multi-View RGB Videos for 3D Dynamic Reconstruction** Zurich, Switzerland
ETH Zurich - AIT Lab
March 2025 - July 2025
 - This semester project builds upon the [GauSTAR framework](#) that focuses on 3D reconstruction for general dynamic objects using multi-view input. The primary objective of this project is to remove the dependence of GauSTAR on depth sensor input; using a novel, RGB-only scene flow estimation method while preserving the performance of the 3D reconstruction
 - **Technical Skills:** Python, PyTorch, 3D Gaussian Splatting
- **Warping Voxel-grids for 3D Robotic Mapping** Zurich, Switzerland
ETH Zurich - 3D Vision course
March 2025 - July 2025
 - This course project under supervision of Dr. Daniel Barath aims to integrate loop closure optimization into voxel-based 3D mapping by developing a method to warp the voxel grid in response to updated camera trajectories.
 - **Technical Skills:** C++, Python, PyTorch
- **Impact of Network Coding on Age of Information in IoT Networks** Istanbul, Turkey
Bogazici University
September 2023-June 2024
 - Completed the senior design project as an undergraduate research fellowship student in this TUBITAK (The Scientific and Technological Research Institution of Turkey) funded project under the supervision of Prof. Mutlu Koca.
 - Designed an energy harvesting network with network coded multi-casting scheme for future use in the later stages of the project.
 - **Technical Skills:** MATLAB

RESEARCH EXPERIENCE

- **Visiting Researcher (MIT Affiliate)** Boston, MA
Broad Institute of MIT and Harvard - Traverso Lab *July 2026 – Current*
 - Conducting research at the intersection of artificial intelligence, computer vision, augmented reality, and biomedical applications.
 - Designing and developing intelligent systems for perception, reasoning, and human interaction in laboratory environments.
 - Collaborating with researchers from the Broad Institute and ETH Zürich on research projects spanning computer vision, machine learning, and digital health technologies.
 - **Technical Skills:**
Python, Computer Vision, Machine Learning, AR/VR, VisionOS, PyTorch, LLMs
- **Undergraduate Research Assistant** Istanbul, Turkey
Bogazici University *December 2021 - July 2024*
 - Took part in an industrial anomaly detection research project with a focus on guided diffusion models, under the supervision of Prof. Ayşın Baytan Ertüzün.
 - Carried out the implementation of independent component analysis through the FastICA algorithm on plant leaf images for the “Signal Processing and Artificial Intelligence Methods for Smart Agriculture” research project.
 - Carried out the implementation of particle swarm optimization methods into independent component analysis using Python.
 - **Technical Skills:** Python, PyTorch, MATLAB, Git.

INDUSTRY EXPERIENCE

- **Image Processing Intern** Ankara, Turkey
ASELSAN A.Ş. *August 2023 - September 2023*
 - Carried out multi-vision geometry simulations and camera calibration for robotics applications.
 - Reviewed image formation and camera models to write an extensive report for future training uses of the department.
 - Carried out Python implementations of the mentioned subjects for several camera setups. Documented and presented the results for future use in the company’s robotics projects.
 - **Technical Skills:** Python, Ubuntu, LaTeX
- **Software Test and Verification Intern** Istanbul, Turkey
Arçelik A.Ş. *July 2022 - September 2022*
 - Designed automated test scenarios for smart IoT devices and the official smart home mobile application of Arçelik.
 - Designed scenarios according to behavior-driven development (BDD) techniques for optimum testing results and ease of collaboration with other teams.
 - **Technical Skills:** Java, Cucumber, Appium, Android Studio, Node.js

SKILLS SUMMARY

- **Research Areas:** Computer vision, machine learning, 3D vision, multi-view geometry, AR/VR, signal processing, Robotics & Control
- **Programming:** Python, C++, C, MATLAB, Swift
- **Frameworks & Tools:** PyTorch, OpenCV, Git, visionOS
- **Languages:** English (Advanced, TOEFL Score: 113/120), Spanish (Intermediate), Turkish (Native)

HONORS & AWARDS

- **Heyning-Roelli Foundation Scholarship, 2026:** Competitive scholarship supporting Master’s thesis research exchange at the Broad Institute of MIT and Harvard.
- **TÜBİTAK Undergraduate Research Fellowship, 2023–2024:** Scholarship supporting Bachelor’s senior design project.